The Effect of Position on the Percentage of Body Mass Supported During Traditional and Modified Push-up Variants

David N. Suprak,[1] Jay Dawes,[2] and Mark D. Stephenson[2]

[1]*Department of Physical Education, Health, and Recreation, Western Washington University, Bellingham, Washington; and*
[2]*Naval Special Warfare Development Group, Virginia Beach, Virginia*

Abstract

Suprak, DN, Dawes, J, and Stephenson, MD. The effect of position on the percentage of body mass supported during traditional and modified push-up variants. *J Strength Cond Res* 24(x): 000–000, 2010—The push-up is a popular upper-extremity weight-bearing exercise. However, limited information is available regarding its effectiveness. Much of the past research has focused on muscle activation levels, whereas very little has examined the forces encountered during push-up variants. The purpose of the present study was to examine the effect of position within the range of motion on the percentage of body mass (BM) supported by the upper extremities during the traditional and modified (knees-down) push-up. Twenty-eight highly strength-trained male subjects were positioned with their hands on a force platform in 4 static positions, consisting of the up and down position in both the traditional and modified push-up exercise. The performance measures included the average vertical ground reaction force (GRF), expressed as a percentage of BM, supported in each of the 4 static positions and the percentage of change between the up and down positions in each push-up exercise. In both the traditional and modified push-ups, subjects supported less weight in the up vs. the down position. The percentage change in % BM from the up to the down position was greater in the modified push-up variant. The pattern of resistances to the push-up exercises observed in this study may be a result of differing moment arms between the support surface contact point (knees or feet) and the hands. These results may be useful in prescribing programs for strengthening and/or rehabilitation for both the prime movers and stabilizers of the upper extremity. Further, range of motion may need to be altered to accommodate strength differences in beginners and clients rehabilitating from injury.

Key Words upper extremity, kinetics, closed kinetic chain, muscle activation

Introduction

Upper-extremity weight-bearing exercises are often utilized in programs designed for strengthening and rehabilitation. The push-up exercise and its many variations are popular in this regard because of their convenience and easy adaptability to various difficulty levels (22) and their theorized tendency to improve joint stability and proprioception during the exercise as a result of compression forces, a feature especially important at the shoulder with its lack of inherent stability (18). The push-up has been used extensively in the testing (9,16,24) and strength training (12,15,28,31) programs of many diverse populations, including tactical and sports athletes and children. It is commonly used with athletes as part of a dynamic warm-up and as an alternative to traditional weight training methods, such as using manual overload when equipment may be scarce. The push-up is also commonly used in rehabilitation of upper-extremity injury, especially in the shoulder (23). It is surprising that, despite the popularity of the push-up as a strengthening and rehabilitation exercise, limited information is available in the literature regarding its potential effectiveness. When developing any periodized training program, the intensity and total volume of training are important contributors to the adaptations achieved. Presently, the only quantifiable information used to identify training volume for the push-up consists of multiplying the sets by the repetitions completed. A better understanding of the training load encountered during the exercise can aid the strength and conditioning professional, clinician, and/or trainee in making better predictions regarding the optimal amount of volume and intensity selected and enable more appropriate adjustments to training programs based on this information.

Much of the research regarding the potential of the push-up exercise to bring about positive adaptations has, to

Address correspondence to David N. Suprak, David.Suprak@wwu.edu.

0(0)/1–7

Journal of Strength and Conditioning Research
© 2010 National Strength and Conditioning Association

Copyright © Lippincott Williams & Wilkins. Unauthorized reproduction of this article is prohibited.

date, centered on the muscle activation patterns of the upper extremity (2,4–6,10,11,13,18–21,23,26,30,32,34,35) and trunk (1,8,14). These authors have reported that overall muscle activation of the prime movers increases as variants of the exercise become more intense (2,10,32). In addition, the pattern of activation in the upper-extremity synergists and the scapular and trunk stabilizer muscles depends on hand position and joint angles during the execution of the movement (presumably as a result of changes in muscle lengths and moment arms) and the support surface used (1,2,4–6,8,11,14,18,21,23,25,26,30,34,35).

Few authors have reported data regarding the forces encountered during variations of the push-up exercise, although they are important determinants of the muscle activation levels observed (7,10,22). Lou reported the greatest posterior and varus elbow shear forces while performing a push-up with the forearm in 90 degrees of internal rotation (22). Donkers reported elbow axial forces during the traditional push-up in a range of 37 to 45% of BM (7). To our knowledge, Gouvali and Boudolos (10) are the only authors to report ground reaction forces (GRFs) observed during the execution of push-up variations, an indication of the gravitational resistance to the movement. They indicated that the peak GRF over the full range of motion (ROM) in the traditional push-up was significantly higher than that in the knees-down variation. However, to date, no data are available regarding the GRF observed at points within the ROM of these exercises. This is an important area of study, considering the effect that joint angle and ROM have on muscle lengths, moment arms, and activation levels. Therefore, the purpose of the present study was to examine the effect of position within the range of motion on the percentage of BM supported by the upper extremities during the traditional and modified (knees-down) push-up variants. We hypothesized that the percentage of BM supported in the traditional push-up would be greater than that in the modified push-up as a result of the greater moment arm of the body center of mass caused by its increased distance from the point of contact with the ground in this exercise. We further hypothesized that in both the traditional and modified push-up, the percentage of BM supported would be greater in the down position compared to the up position for the same reason. The knowledge gained from the results of this study can potentially help in understanding and predicting the muscle activation and loading patterns previously reported and likely adaptations to training with these exercises.

Methods

Experimental Approach to the Problem

Given the paucity of data concerning the forces encountered during the performance of the push-up exercise and its variants, this study was undertaken to examine the differences in the percentage of BM that was supported in 2 positions (up and down) within the ROM of the traditional and modified (on-knees) variations of the push-up exercises. All testing was performed in a single session for each subject, with ample recovery between testing of each position to minimize the effects of fatigue. Subjects were asked to assume a position consisting of rigid spine and lower-extremity segments during data collection in each position of each push-up variant. The independent variables involved were the type of push-up variant (traditional vs. modified) and the position within the ROM (up vs. down). The dependent variables for this study were the percentage of the BM supported in each position (up and down) under each condition (traditional and modified push-up) and the percent change from the up to the down position in each variation of the push-up.

Subjects

Twenty-eight (28) male subjects (age 33.62 ± 8.59 years, BM 84.66 ± 11.65 kg, height 178.98 ± 5.76 cm) participated in this study. These individuals were highly strength trained (currently involved in resistance training 3 to 4 days per week), were members of special forces and SWAT units training at the National Strength and Conditioning Association World Headquarters, and were experienced in the push-up techniques used in this study. Subjects were healthy, with no history of upper-extremity or spine pathology. This experiment was approved by the institutional review board of the University of Colorado at Colorado Springs for the protection of human subjects, and all subjects were informed of the procedures and potential risks involved before they gave their written consent to participate.

Procedures

Data collection was performed using an AMTI force platform (Newton, Massachusetts, USA) sampling at 200 Hz and interfaced with AccuPower software (Athletic Republic, Grand Rapids, MI, USA). Prior to testing, each subject was weighed on the force platform. Subjects were then instructed to position their hands slightly wider than shoulder-width apart and in the approximate center of the force plate (indicated by a strip of tape placed on the force platform), with the shoulders positioned over the hands (Figure 1). While maintaining an angle of approximately 180 degrees between the upper and lower body, subjects were instructed to maintain 4 different isometric positions: a traditional push-up position with the arms fully extended (Figure 1A), a traditional push-up position with the elbows flexed at approximately 90 degrees (Figure 1B), a modified push-up position with the arms fully extended (Figure 1C), and a modified push-up position with the elbows flexed at approximately 90 degrees (Figure 1D). All positions were held for 6 seconds during data collection. Three trials were collected in each condition. Subjects were instructed to wait as long as necessary, but at least 1 minute, between trials to minimize the effects of fatigue on the ability to perform the task. The order of testing of each condition was randomly assigned for each subject. For each subject, mean vertical forces were calculated and averaged over the 3 trials in each position and then expressed as a percentage of the subject's BM. These mean vertical forces were then averaged across the 28 subjects, and this mean value was used for all subsequent

Copyright © Lippincott Williams & Wilkins. Unauthorized reproduction of this article is prohibited.

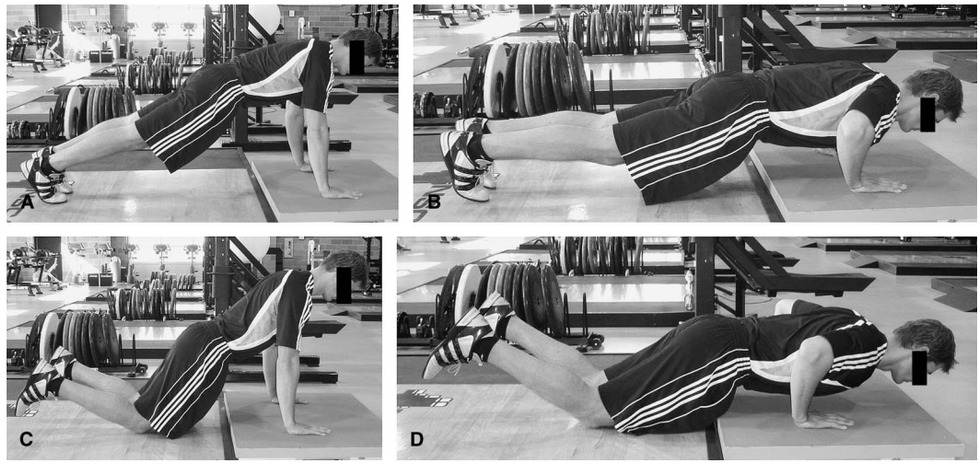


Figure 1. Experimental set-up for traditional push-up, up position (A); traditional push-up, down position (B); modified push-up, up position (C); and modified push-up, down position (D).

analyses. In addition, the percent change observed from the up to the down position in both the traditional and modified exercises was calculated as the difference between the percentage of BM supported in the up and down position divided by the percentage supported in the up position. No nutritional or hydration control was implemented in this study because all measures were within subjects in 1 testing session.

Statistical Analyses

Statistical analysis was performed using SPSS, version 16 (Chicago, Illinois, USA). Effect sizes were calculated between the up and down positions of the push-up variants (down position minus up position divided by the pooled standard deviation of both positions) and between the modified and traditional variants across the 2 positions (traditional minus modified divided by the pooled standard deviation of both variants), along with the percentage change in both variants (modified minus traditional percentage change divided by the pooled standard deviation of both variants). A 2-way repeated-measures analysis of variance (ANOVA) was used to determine significant changes in the mean percentage of BM supported in the up and the down position in both the traditional and modified push-up. Significant interaction effects were followed by simple effect analyses using *t*-tests. A paired samples *t*-test was used to determine differences in the percent change from the up to the down position in the traditional and modified push-up variations. Following a Bonferroni correction for multiple tests, the alpha level was set at $p \leq 0.025$ to determine significant effects in all analyses.

Results

A power analysis was conducted using a minimal detectable difference between conditions of 5% BM, a standard deviation of 3.3%, and an alpha level of 0.05. According to this analysis, the statistical power associated with 28 subjects was 0.98. The measurement showed good reliability, with an intraclass correlation coefficient for the percentage of BM supported across the 4 conditions of 0.848. In the traditional push-up, subjects supported 69.16% (±2.83%) of their BM in the up position, and 75.04% (±2.62%) in the down position. In the modified push-up, subjects supported 53.56% (±4.27%) of their BM in the up position and 61.80% (± 3.48%) in the down position. The 2-way repeated measures ANOVA revealed a significant type by position interaction ($F[1,27] = 18.42$, $p < 0.001$) (Figure 2). Therefore, the main effects of type and position were not examined. An analysis of simple effects

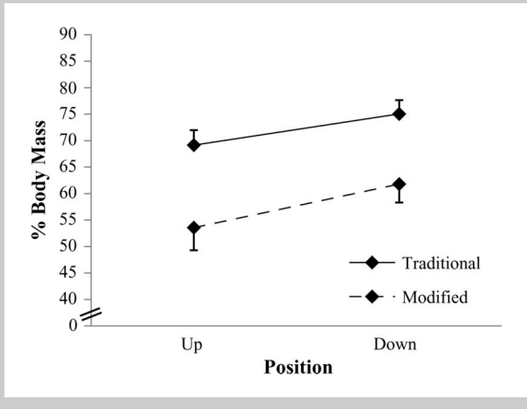


Figure 2. Mean (± *SD*) percentage of body mass supported in the up and down positions during the traditional and modified push-up variants.

Copyright © Lippincott Williams & Wilkins. Unauthorized reproduction of this article is prohibited.

Table 1. Effect sizes for statistical analyses. Pooled *SD* was used for all calculations.

Comparison	Effect size
Down vs. up position	0.81
Traditional vs. modified variant	1.65
Traditional vs. modified % change	1.08

revealed that a significantly greater proportion of BM was supported in the down position than in the up position in both the traditional ($F[1,27] = 24276.07$, $p < 0.001$) and modified ($F[1,27] = 7493.69$, $p < 0.001$) push-up variants. Simple effect analysis also revealed that a significantly greater proportion of BM was supported in the traditional than in the modified variant in both the up ($F[1,27] = 10078.30$, $p < 0.001$) and down ($F[1,27] = 15985.97$, $p < 0.001$) positions. The percent change in the BM supported from the up to the down position in the traditional and modified push-ups was 8.59 ± 3.77% and 15.76 ± 7.03%, respectively. The paired *t*-test revealed that the percent change in the proportion of BM supported in the up and down position was significantly greater in the modified push-up than in the traditional push-up variation ($t[27] = 6.40$, $p < 0.001$). According to Cohen's standards, all effect sizes calculated in this study were large (3) (Table 1).

Discussion

The purpose of the present study was to determine the effect of the position within the push-up ROM on the percentage of BM supported in both the traditional and modified (knees-down) push-up variations. We hypothesized that, because of the greater moment arm involved, the percentage of BM supported would be greater in the traditional than in the modified push-up. This hypothesis was supported, as evidenced by the significant simple effect analyses indicating that the percentage of BM was greater in both the up and down positions in the traditional vs. the modified push-up variant. It is difficult to make direct comparisons between this and previous studies in regards to GRFs in various positions in the push-up because we are the first to report such data. However, the overall trend of our results is in agreement with other studies of the kinetics involved in push-up variations. Gouvali and Boudolos (10) reported peak vertical GRFs equal to 66 and 53% BM over the traditional and modified push-up ROM, respectively. These percentages are lower than those observed in the present study. The differences in the percentage of BM supported in the traditional and modified variations between Gouvali's data and that in the present study may be related to a number of factors. Gouvali and Boudolos studied young (age 20 years) recreationally trained men. The subjects in our study were older (age 34 years) and at a higher training status. These factors may have resulted in a difference in the distribution of the mass within the bodies of subjects in the 2 studies, contributing to the disparities observed. Also, Gouvali and Boudolos examined peak GRFs over the ROM of a dynamic push-up movement, whereas we examined GRFs at discrete positions within the ROM. Nevertheless, the results reported by Gouvali and Boudolos are in overall agreement with ours regarding the difference in the resistance encountered during traditional and modified variants of the exercise.

The second hypothesis tested in this study was that a greater percentage of BM would be supported in the down vs. the up position. This hypothesis was also supported, as demonstrated by the significant simple effect analyses indicating that the percentage of BM supported was greater in both the traditional and modified exercises in the down position as compared to the up position. Donkers et al. (7) reported axial elbow joint reaction forces of 37 and 38% BW in the up and down positions of the traditional push-up, respectively. Although it is difficult to equate elbow joint reaction forces to GRFs examined in this study, it is evident that the trend of changing forces reported by Donkers in the up and down positions of the traditional push-up is consistent with that observed in the present study. We hypothesize that this pattern of increased force in the down position is the consequence of the whole-body center of mass being located further from the point of contact (feet or knees) with the support surface (floor), resulting in greater torque that must be matched by the force exerted against the force plate. This would also be the rationale explaining the greater force exerted against the force plate in the traditional push-up compared with that in the modified version.

The implications of these results are better understood when considered in the context of studies on muscle activation patterns during various forms of the push-up exercise. Several authors examining the activation patterns of prime movers during variants of the push-up have reported increased electromyograpy (EMG) levels under conditions of increased demands (10,25,32,35). Gouvali and Boudolos (10) reported that the traditional push-up resulted in greater EMG activity in both the pectoralis major and the triceps brachii muscles than did the modified push-up. Uhl et al. (35) examined EMG activity of the pectoralis major, anterior deltoid, posterior deltoid, and infraspinatus during 7 different positions of increasing demand on the upper extremity, including prayer position, quadruped, tripod, pointer, traditional push-up (up position), push-up with feet elevated, and 1-arm push-up. They indicated that EMG levels in all muscles examined increased with increasing upper-extremity weight-bearing. Both the pectoralis major (32) and the triceps brachii (21,25) muscles have also been reported to be more active during a push-up on a Swiss ball compared to a traditional push-up, again underscoring the effect of task difficulty on muscular demand. Because muscular demand in

Copyright © Lippincott Williams & Wilkins. Unauthorized reproduction of this article is prohibited.

the prime movers appears to increase with increased weight-bearing in the upper extremity, we hypothesize that muscle activation would be higher in the traditional vs. the modified push-up and would increase from the up to the down position in both variations. However, because EMG was not recorded in the present study, this hypothesis remains speculative and should be examined directly in the future.

Scapular and humeral stabilizers have received some attention in the EMG literature during push-up variants over a range of difficulty (4–6,23,35), owing to their role in maintenance of proper positioning and scapulohumeral rhythm during arm elevation and the specific focus placed on their function in the rehabilitation setting (23). The modified push-up and the traditional push-up with a plus, which includes scapular protraction at the top of the push-up ROM, appear to preferentially activate the serratus anterior with respect to the upper trapezius, a feature important for proper positioning of the scapula during arm movement (5,23). Additionally, performing the push-up with a plus also results in relatively high activation of both the upper and lower subscapularis (6) and infraspinatus (35) muscles. In contrast, performance of the wall push-up exercise results in a high upper trapezius/serratus anterior ratio (4). Therefore, the degree of activation of the various scapular and humeral stabilizers during upper-extremity exercises may be dependent on a complex interplay between the weight-bearing demand of the exercise and the degree of arm elevation during execution. The modified push-up is characterized by a lower arm elevation level and a smaller weight-bearing demand but equivalent serratus anterior activation, as compared to the push-up with a plus exercise (5). The wall push-up is less demanding, involves approximately the same arm elevation angle, and resulted in a higher upper trapezius/serratus anterior ratio compared with that in the traditional push-up (4). Therefore, increased weight-bearing in the upper extremities results in heightened activity of the supraspinatus and infraspinatus muscles, presumably because of the increased need for glenohumeral stabilization under conditions of increasing forces, as evidenced by the data in the present study. Also, as arm elevation increases during an exercise, greater demand may be placed on the upper trapezius than can be countered by the serratus anterior, unless the weight-bearing role of the upper extremities during the exercise is increased as well. Thus, incorporating a traditional push-up exercise into a training program may be especially helpful in strengthening the rotator cuff muscles, whereas performing a modified push-up may result more in strengthening of the serratus anterior muscle and optimizing the upper trapezius/serratus anterior ratio during upper-extremity rehabilitation programs.

It is also important when interpreting the present results to consider the pattern of human strength during the pressing motion involved in the traditional and modified push-up. Research examining human strength curves in upper-extremity pressing motions has focused on the bench press exercise, a movement that is biomechanically similar to the push-up (17,27,29,36). During the bench press exercise, the lowest force is exerted at the bottom of the lift, in a position of shoulder horizontal abduction and elbow flexion (the sticking point) (36), and the highest force is exerted near full extension of the elbows (27), an angle of approximately 120 degrees (29). This pattern of strength expression is termed an ascending strength curve (33). The results of the present study indicate that the resistance encountered in both the traditional and modified push-up variants exhibits a pattern opposite that for human strength expression in the bench press upper-body pressing motion. Whereas strength would likely be greatest in the up position of both push-up variations, and lowest in the down position, the percentage of BW supported in these positions showed the opposite trend. This difference in strength and load patterns is important to consider when designing a conditioning program for individuals beginning a body weight resistance training program and for individuals rehabilitating an upper-extremity injury and beginning closed kinetic chain exercises. In either case, it may be necessary to modify the ROM of the push-up exercise to accommodate differences in strength and to progressively increase the ROM as the individual increases in strength.

To gain more insight into the trend of changing forces during the ROM of the 2 push-up variants studied here, an analysis was performed to assess the difference between the traditional and modified push-up with regard to the proportional amount of increase in the percentage of BM supported from the up to the down position. The paired *t*-test indicated a significantly greater increase in the percentage of BM supported in the modified than in the traditional push-up variant. This difference is presumably a result of the greater angle created between the support surface and the torso in the up position and the greater increase in the moment arm of the center of mass (COM), in the modified vs. the traditional push-up. This finding may have important implications for the prescription of strengthening programs, especially for those individuals with shoulder or lumbar spine pathologies. When prescribing strengthening programs incorporating the modified push-up exercise, especially for those just beginning a resistance training program or for those rehabilitating a shoulder or lumbar spine injury, it is important to be cognizant of the relatively greater change in resistance over the full ROM. Particularly in the early stages of closed chain exercises in a post-surgical shoulder rehabilitation program, the demand placed on the serratus anterior muscle at the bottom of the ROM in the modified push-up may be too great and may promote altered scapular kinematics. This supposition warrants further direct investigation. In the same way, although overall the modified push-up presents a lower-intensity exercise compared with the traditional variation, it is important to monitor the trainee to ensure correct execution of the exercise, especially at the bottom of the ROM.

Copyright © Lippincott Williams & Wilkins. Unauthorized reproduction of this article is prohibited.

There are several limitations to the results of this study. The data are derived from only 2 static positions within 2 push-up variants. Forces are different during dynamic movement and static positioning. Also, forces will likely vary over the entire ROM of a given exercise. Therefore, future studies should examine performance of these exercises over the entire ROM. This study only observed vertical GRFs at the various positions. It would be interesting to examine the anteroposterior and mediolateral shear forces during the push-up variations to gain insight into possible forces acting at proximal joints for injury prevention purposes. Also, no EMG was collected in the present study, so no information is available regarding muscle activation patterns at the various positions. Future investigations should include 3-dimensional force and EMG measurements over the ROM of the exercises to develop a deeper understanding of the demands of the exercises and their potential for improvement of upper-extremity function and their utility in programs for upper-extremity rehabilitation. Last, the subjects in this study were highly resistance-trained males. Future studies should include comparisons of forces and muscle activation patterns of trained and untrained males and females during traditional and modified push-ups, considering gender differences in body mass distribution and the adaptations in muscle mass occurring with resistance training.

Practical Applications

According to the principle of specificity, use of the traditional or modified push-up in a program for strengthening or injury prevention or rehabilitation in the upper extremity depends on the specific objective of the program. The modified push-up is an effective alternative for individuals unable to perform a traditional push-up and appears to be the option of choice for those beginning a weight-bearing rehabilitation program aimed at optimizing the strength ratio between the upper trapezius and serratus anterior, given the reduced percentage of BW supported and elevated serratus anterior activity during the exercise. The traditional push-up is more demanding than the modified push-up and may be more effective in strengthening the prime movers and rotator cuff muscles because of increased forces applied. Rehabilitation clinicians may, therefore, elect to utilize the modified push-up to promote optimal scapulothoracic stability in the early phases of weight-bearing shoulder rehabilitation and to emphasize the traditional push-up as the rehabilitation program progresses to focus on regaining glenohumeral joint stability and strength. Likewise, the strength and conditioning professional may utilize the modified push-up to promote shoulder girdle stability and the traditional push-up to develop rotator cuff and shoulder prime mover strength and endurance, especially in individuals in the early phases of training, as a dynamic warm-up to prepare for more demanding work in advance athletes and as an adjunct to more traditional resistance training methods.

When incorporating a traditional or modified push-up into rehabilitation or reconditioning programs for patients with upper-extremity injuries or for individuals being introduced to resistance training, the ROM or exercise may need to be altered so that it can be performed safely and so the increase in resistance through the ROM does not put undue stress on any weak or injured structure(s), keeping in mind that, although the overall resistance is lower in a modified push-up compared with a traditional push-up, the change in resistance is greater over the ROM.

References

1. Beach, TA, Howarth, SJ, and Callaghan, JP. Muscular contribution to low-back loading and stiffness during standard and suspended push-ups. *Hum Mov Sci* 27: 457–472, 2008.
2. Cogley, RM, Archambault, TA, Fibeger, JF, Koverman, MM, Youdas, JW, and Hollman, JH. Comparison of muscle activation using various hand positions during the push-up exercise. *J Strength Cond Res* 19: 628–633, 2005.
3. Cohen, J. *Statistical power analysis for the behavioral sciences*. Hillsdale, NJ: Lawrence Erlbaum Associates, 1988.
4. de Oliveira, AS, de Morais Carvalho, M, and de Brum, DP. Activation of the shoulder and arm muscles during axial load exercises on a stable base of support and on a medicine ball. *J Electromyogr Kinesiol* 18: 472–479, 2008.
5. Decker, MJ, Hintermeister, RA, Faber, KJ, and Hawkins, RJ. Serratus anterior muscle activity during selected rehabilitation exercises. *Am J Sports Med* 27: 784–791, 1999.
6. Decker, MJ, Tokish, JM, Ellis, HB, Torry, MR, and Hawkins, RJ. Subscapularis muscle activity during selected rehabilitation exercises. *Am J Sports Med* 31: 126–134, 2003.
7. Donkers, M, An, KN, Chao, EY, and Morrey, BF. Hand position affects elbow joint load during push-up exercise. *J Biomech* 26: 625–632, 1993.
8. Freeman, S, Karpowicz, A, Gray, J, and McGill, S. Quantifying muscle patterns and spine load during various forms of the push-up. *Med Sci Sports Exerc* 38: 570–577, 2006.
9. Gabbett, TJ, Johns, J, and Riemann, M. Performance changes following training in junior rugby league players. *J Strength Cond Res* 22: 910–917, 2008.
10. Gouvali, MK and Boudolos, K. Dynamic and electromyographical analysis in variants of push-up exercise. *J Strength Cond Res* 19: 146–151, 2005.
11. Hardwick, DH, Beebe, JA, McDonnell, MK, and Lang, CE. A comparison of serratus anterior muscle activation during a wall slide exercise and other traditional exercises. *J Orthop Sports Phys Ther* 36: 903–910, 2006.
12. Harman, EA, Gutekunst, DJ, Frykman, PN, Nindl, BC, Alemany, JA, Mello, RP, and Sharp, MA. Effects of two different eight-week training programs on military physical performance. *J Strength Cond Res* 22: 524–534, 2008.
13. Hinson, MM. An electromyographic study of the push-up for women. *Res Q* 40: 305–311, 1969.
14. Howarth, SJ, Beach, TA, and Callaghan, JP. Abdominal muscles dominate contributions to vertebral joint stiffness during the push-up. *J Appl Biomech* 24: 130–139, 2008.
15. Kilinc, F. An intensive combined training program modulates physical, physiological, biomotoric, and technical parameters in women basketball players. *J Strength Cond Res* 22: 1769–1778, 2008.
16. Knapik, JJ, Rieger, W, Palkoska, F, Van Camp, S, and Darakjy, S. United States Army physical readiness training: Rationale and evaluation of the physical training doctrine. *J Strength Cond Res* 23: 1353–1362, 2009.

Copyright © Lippincott Williams & Wilkins. Unauthorized reproduction of this article is prohibited.

17. Lander, J, Bates, B, Sawhill, J, and Hamill, J. A comparison between free-weight and isokinetic bench pressing. *Medicine and Science in Sports and Exercise* 17: 344–353, 1985.

18. Lear, LJ and Gross, MT. An electromyographical analysis of the scapular stabilizing synergists during a push-up progression. *J Orthop Sports Phys Ther* 28: 146–157, 1998.

19. Lehman, GJ. An unstable support surface is not a sufficient condition for increases in muscle activity during rehabilitation exercise. *JCCA J Can Chiropr Assoc* 51: 139–143, 2007.

20. Lehman, GJ, Gilas, D, and Patel, U. An unstable support surface does not increase scapulothoracic stabilizing muscle activity during push up and push up plus exercises. *Man Ther* 13: 500–506, 2008.

21. Lehman, GJ, MacMillan, B, MacIntyre, I, Chivers, M, and Fluter, M. Shoulder muscle emg activity during push up variations on and off a swiss ball. *Dyn Med* 5: 7, 2006.

22. Lou, S, Lin, CJ, Chou, PH, Chou, YL, and Su, FC. Elbow load during pushup at various forearm rotations. *Clin Biomech (Bristol, Avon)* 16: 408–414, 2001.

23. Ludewig, PM, Hoff, MS, Osowski, EE, Meschke, SA, and Rundquist, PJ. Relative balance of serratus anterior and upper trapezius muscle activity during push-up exercises. *Am J Sports Med* 32: 484–493, 2004.

24. Mangine, GT, Ratamess, NA, Hoffman, JR, Faigenbaum, AD, Kang, J, and Chilakos, A. The effects of combined ballistic and heavy resistance training on maximal lower- and upper-body strength in recreationally trained men. *J Strength Cond Res* 22: 132–139, 2008.

25. Marshall, P and Murphy, B. Changes in muscle activity and perceived exertion during exercises performed on a swiss ball. *Appl Physiol Nutr Metab* 31: 376–383, 2006.

26. Martins, J, Tucci, HT, Andrade, R, Araújo, RC, Bevilaqua-Grossi, D, and Oliveira, AS. Electromyographic amplitude ratio of serratus anterior and upper trapezius muscles during modified push-ups and bench press exercises. *J Strength Cond Res* 22: 477–484, 2008.

27. Massey, CD, Vincent, J, Maneval, M, Moore, M, and Johnson, JT. An analysis of full range of motion vs. Partial range of motion training in the development of strength in untrained men. *J Strength Cond Res* 18: 518–521, 2004.

28. McGuigan, MR, Tatasciore, M, Newton, RU, and Pettigrew, S. Eight weeks of resistance training can significantly alter body composition in children who are overweight or obese. *J Strength Cond Res* 23: 80–85, 2009.

29. Mookerjee, S and Ratamess, N. Comparison of strength differences and joint action durations between full and partial range-of-motion bench press exercise. *J Strength Cond Res* 13: 76–81, 1999.

30. Moseley, JB Jr, Jobe, FW, Pink, M, Perry, J and Tibone, J. Emg analysis of the scapular muscles during a shoulder rehabilitation program. *Am J Sports Med* 20: 128–134, 1992.

31. Roberts, MA, O'Dea, J, Boyce, A, and Mannix, ET. Fitness levels of firefighter recruits before and after a supervised exercise training program. *J Strength Cond Res* 16: 271–277, 2002.

32. Sandhu, JS, Mahajan, S, and Shenoy, S. An electromyographic analysis of shoulder muscle activation during push-up variations on stable and labile surfaces. *Int J Should Surg* 2: 30–35, 2008.

33. Stone, MH, Stone, M, and Sands, WA. *Principles and Practice of Strength Training*. Champaign, IL: Human Kinetics, 2007.

34. Tucker, WS, Campbell, BM, Swartz, EE, and Armstrong, CW. Electromyography of 3 scapular muscles: A comparative analysis of the cuff link device and a standard push-up. *J Athl Train* 43: 464–469, 2008.

35. Uhl, TL, Carver, TJ, Mattacola, CG, Mair, SD, and Nitz, AJ. Shoulder musculature activation during upper extremity weight-bearing exercise. *J Orthop Sports Phys Ther* 33: 109–117, 2003.

36. Wilson, GJ. Bar path and force profile characteristics for maximal and submaximal loads in the bench press. *Int J Sport Biomech* 5: 390–402, 1989.

Copyright © Lippincott Williams & Wilkins. Unauthorized reproduction of this article is prohibited.